

Aryan Singh

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computing and Financial Management (Co-op) | GPA: 3.70 / 4.00

Expected Apr. 2030

- **Relevant Coursework:** Algorithms & Data Structures, Financial Markets, Data Analytics, Financial Reporting

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL (MySQL), HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Tailwind CSS, Streamlit

Data & Machine Learning: pandas, NumPy, scikit-learn, Matplotlib, yfinance, Recharts

Finance & Analytics: Financial Modeling, Portfolio Optimization, Risk Metrics, DCF Valuation

Developer Tools: Git, GitHub, Jupyter Notebook, Power BI, Excel, Vercel, Render

PROJECTS

EPS Drift Scanner | *Python, JavaScript, FastAPI, React, pandas, scikit-learn, Recharts*

[GitHub](#)

- Developed a live EPS surprise scanner tracking **analyst estimates vs. actuals** across 10+ equities, deployed full-stack on Vercel and Render.
- Engineered a **quantitative drift pipeline** classifying quarterly EPS surprises across 5 tiers and 12 quarters with configurable Wall Street estimate alert thresholds.
- Detected anomalous earnings surprises via residual scoring, surfacing **per-equity anomaly scores** on an interactive dashboard to flag outlier reporting events.

Market Stress & Crisis Simulator | *Next.js, TypeScript, React, Tailwind CSS, Recharts*

[GitHub](#)

- Built a **portfolio simulation engine** enabling custom allocation across 10+ assets and 3 major market crises with configurable rebalancing and normalization.
- Engineered a **stress-testing framework** replaying 2008 GFC, COVID-19, and 2022 Rate Shock over 12–36 month horizons, enabling side-by-side drawdown and recovery comparison.
- Designed an **analytics pipeline** processing 10+ years of historical market data to compute 6+ risk metrics – maximum drawdown, Sharpe ratio, and time-to-recovery – delivering quantifiable portfolio insights.

Robo-Advising Portfolio Optimizer | *Python, pandas, NumPy, yfinance*

[GitHub](#)

- Implemented a **portfolio construction pipeline** filtering equities by currency, liquidity, and market capitalization, removing delisted tickers and resolving cross-listed duplicates across 10+ industries.
- Screened equities by **annualized volatility** to exclude high-risk names per sector while preserving diversification, reducing overall portfolio risk exposure.
- Built a **\$1,000,000 CAD market-meet portfolio** correlated with the S&P 500 and TSX Composite, converting weights into shares net of FX and transaction costs, producing a fully investable allocation.

Equity Trend Analyzer | *Python, pandas, NumPy, yfinance, Matplotlib, Streamlit*

[GitHub](#)

- Constructed a **quantitative equity analysis tool** computing maximum drawdown, total return, and annualized volatility across 50+ equities from historical market data.
- Classified trend direction (**uptrend/downtrend/neutral**) via log-price regression, identifying momentum shifts for early trend detection.
- Engineered **RSI (14)** and moving average crossover signals with dynamic visualizations and CSV export, enabling downstream quantitative analysis workflows.

EXPERIENCE

Food & Beverage Service Associate

Mar. 2023 – Aug. 2025

Canada's Wonderland

Vaughan, ON

- Processed **300+ high-volume guest transactions** per shift using Oracle POS systems, maintaining **95–99%** accuracy during peak operating hours.
- Coordinated operations across **5–10 associates per shift**, delegating tasks to cut wait times and sustain service standards during peak periods.

International Service Project (Global Development)

Mar. 2024

Shelter Them Poverty Relief (NGO) — Brampton Christian School

Kigali, Rwanda

- Planned and executed fundraising initiatives, raising **\$2,000+** and coordinating budget allocation to support on-time project delivery.
- Collaborated across a cross-functional team of **15+ members** to deliver a community infrastructure project on time and under budget despite logistical constraints.